

Bandage to automatically deliver medicine to a wound

Researchers from MIT have developed a new type of bandage that incorporates electronics and drug reservoirs to monitor and care for a wound. The durable hydrogel bandage is supple and flexible, allowing for full range of movement even if it is applied to a knee or an elbow. It also has a few innovative features such as temperature-monitoring electronics that automatically release medicine to fight infections.



Smart bandage has embedded electronics to track and treat wounds (Image courtesy: www.popularmechanics.com)

The smart wound dressing is made of a rubbery hydrogel matrix that is 90 per cent water, one designed specifically to replicate the qualities of human tissue. The gel creates a strong bond with materials such as titanium, aluminium, silicon, ceramic, gold and other substances that are commonly used to build electronics.

Titanium wire runs through the gel to make the bandage conductive, allowing a number of electronic devices to be embedded, such as semiconductor chips. LED lights are also used; these can flash when a wound reaches a certain temperature or drug reservoirs run low. Medicine reservoirs are drilled into the hydrogel and travel to the wound via channels cut in the matrix.

Wireless, dissolvable sensors to monitor brain

A team of neurosurgeons and engineers has developed wireless brain sensors that monitor pressure and temperature inside the brain and are then absorbed by the body, so that there is no need for surgery to remove the devices.

Developed by scientists at Washington University School of Medicine in St. Louis and engineers at University of Illinois at Urbana-Champaign, the implants can be used to monitor patients with traumatic brain injuries.

The devices are made mainly of polylactic-co-glycolic acid and silicone and can transmit accurate pressure and temperature readings as well as other information.

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